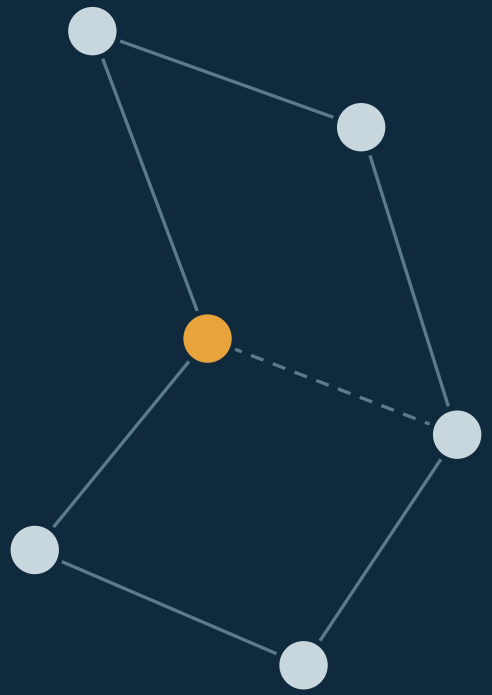
IVEXA

Computing and trust for devices that cannot rely on the network — or on each other



Patent pending · 3 US provisional applications · 2026

Andrei Zagainov — inventor and sole owner
Non-confidential summary

Three things break distributed systems at the edge



Links drop

Factories, vehicles, ships, remote sites and disaster zones have intermittent connectivity. Most distributed systems then stop or lose finished work.



Devices cannot be taken on trust

A stolen, hacked or faulty device can return wrong results or spend the same digital money twice while offline.



The centre is a single point of failure

When the coordinator or the central server is lost, the whole computation stops.

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IVEXA solves all three in one architecture

Trust and compute infrastructure for devices that cannot rely on the network and cannot take each other at their word.

Gain-aware hybrid computing

US 64/182,856. Servers and ordinary computers work as one system; jobs are distributed only when it measurably pays off.

Computing over unreliable links

US 64/183,524. Work survives outages, untrusted devices are checked blindly, a lost coordinator is replaced at once.

Offline trust and payments

US 64/183,551. Tells a bad link from deliberate blocking; double spending is caught offline by blind witnesses.

Why this technology is worth buying

Solves a costly problem

Outages, lost work and untrusted devices cost money in every distributed system — IVEXA removes these losses.

Uses hardware you already own

Idle computers become extra computing power: fewer new servers and lower cloud bills.

Works where others stop

No internet, poor links, a lost central server — the work and the trust keep going.

Tested and protected

Working code, 73 tests, real-machine results; 3 US patent applications filed, priority from October 2026.

Simple to acquire

One inventor and one owner — a licence or purchase without co-owners or corporate approvals.

What each invention gives your business (1/2)

Hybrid computing with measured gain

Turns the computers a company already owns into extra computing power — and uses them only when it really speeds the work up.

Gives: lower hardware and cloud costs; sensitive data never leaves the company.

Possible uses:

- Private AI in banks, hospitals and law firms
- Universities and research labs
- Engineering simulation and 3D rendering
- Software build and testing farms
- Store and branch computers of retail chains
- Small on-site data centres

Computing over unreliable links

Work keeps going and nothing is lost when the connection drops or a device fails; a faulty or hacked device cannot slip in wrong results.

Gives: continuity, no rework, results you can trust.

Possible uses:

- Ships, aircraft, trains and truck fleets
- Mines, oil and gas sites, wind and solar farms
- Remote clinics and field offices
- Satellite and rural networks
- Smart factories and warehouses
- Emergency services and disaster zones

What each invention gives your business (2/2)

Trust and payments without connection

Devices can trust each other and accept payments safely with no connection, and notice when someone hides that a device was stolen.

Gives: safe offline transactions and fewer fraud losses.

Possible uses:

- Offline card and digital-currency payments
- Transit tickets, parking, tolls and vending
- Payments on planes, ships, festivals, rural areas
- Emergency cash aid when networks are down
- Access control and digital identity
- Revoking stolen or hacked IoT devices

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Technical part

For engineers and specialists: how it works and what the tests show

1 · Compute where it pays off

- Joins servers — even ones busy running an AI model — with ordinary laptops and workstations into one compute system
- Measures each node's speed and each link's delay, predicts the result and distributes jobs only if the gain is real
- Splits work in proportion to measured speed; every result is verified by checksum
- No open inbound ports on user computers; works fully without internet

×1.85

Live test over the internet: loaded server + ordinary laptop, 800 ms link — results identical

×0.87

Small jobs sent over the same link ran slower — the reason IVEXA measures before it distributes

2 · Work that survives broken links and bad devices

100 %

of finished work kept when links drop on every node (emulation)

0 / 1000

jobs lost when the coordinator is destroyed; ~0.1 s delay

0

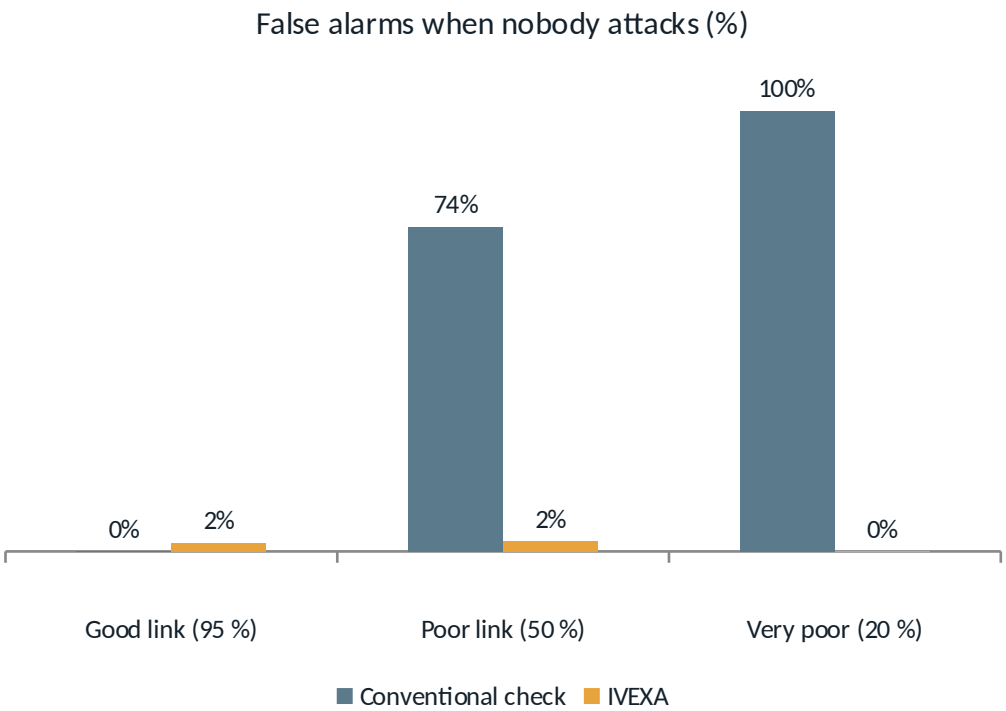
honest devices falsely accused; cheaters caught in 97.5–100 % of cases

×4.46

vs one process: 5 real machines, 8 nodes, real AI model

- When the link drops, a device keeps computing and delivers the result later; waiting time adapts to the real history of outages and runs on the device's own clock, not GPS
- Devices without a hardware security chip (TPM) are double-checked blindly by a second device; disputes are settled by a TPM device; a cheater stays in quarantine until the owner confirms identity
- Real-network test: 16 link cuts during work — 100 % of work kept, results identical, no extra time

3 · Trust and payments without connectivity



- A device tells a simply bad link from someone deliberately blocking revocation messages (e.g. for a stolen device) — and only then tightens checks on neighbours
- Full blocking detected in 100 % of runs within ~3 periods; on a poor link in 99.7 % (~9 periods)
- Offline payments: neighbours blindly witness each transfer, so a double spend is exposed inside a disconnected group within 3 rounds — even with 60 % of witnesses bribed
- Emulation, 300 runs per scenario

Evidence — and what is still emulation

Tested on real machines

- Filing 1: live test over the public internet
- Filing 2: 5 separate machines, 8 nodes, real AI model, degraded links
- 73 automated tests, all passing
- Every result reproducible with one command

Emulated so far

- Filing 3 (offline trust): emulation on one machine; signatures and attestation simplified
- Large fault scenarios of filing 2: emulation with real TCP and encryption
- Next step: field tests on a partner's hardware

Intellectual property

Filing	US App. No.	Subject
23 Oct 2026	64/182,856	Gain-aware distribution across a hybrid network of servers and ordinary computers
24 Oct 2026	64/183,524	Computation over disconnected, intermittent and limited links; blind double-checking; coordinator failover
24 Oct 2026	64/183,551	Detection of suppressed revocation messages and offline blind witnesses

- Sole inventor and applicant: one owner, no company and no co-owners
- Non-provisional deadline: 23 October 2027
- Open to an exclusive or non-exclusive licence (by field of use) or assignment of the portfolio
- Code, tests and full test reports available under NDA
- Further improvements, already developed and tested but not yet filed, can be discussed under NDA

About the inventor and next steps

- Andrei Zagainov, independent inventor
- Author and architect of IVEXA and of all three inventions
- Law degree (Kemerovo State University, 2011); CEO of a software company 2020–2023
- Ready to keep developing the technology with a partner, or to sell or license it

Next steps

1. Introductory video call
2. Mutual NDA
3. Technical deep-dive: code, tests, reports
4. Pilot, licence or partnership

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